COMPUTER FUNDAMENTALS AND INTRODUCTION TO PYTHON

Class 1 Notes for Pharmacy Students

Learning Objectives

- Understand basic computer concepts and the Input–Processing–Output model.
- Differentiate data and information, hardware and software, and system and application software.
- Explain why computers are important in modern Pharmacy.
- Explain programming, programs, programming languages and Python.
- Explain why Python is useful for Pharmacy students.
- Write simple Python programs using print(), input(), variables and arithmetic operators.
- Use int() and float() for simple calculations.

PART A – COMPUTER FUNDAMENTALS

1. What is a Computer?

A computer is an electronic device that accepts data as input, processes it according to instructions, stores information, and produces useful output.

Basic model: Input → Processing → Output

Pharmacy example: Price ₹25 × quantity 10 → total cost ₹250.

2. Why are Computers Important in Pharmacy?

- Pharmaceutical calculations: dose, concentration, dilution, percentage and unit conversion.
- Laboratory work: recording observations, calculating results and preparing reports.
- Drug research: drug discovery, molecular analysis, computational chemistry and screening.
- Clinical research: clinical-trial and patient-data analysis.
- Pharmacokinetics: analysis of drug concentration, time and half-life.
- Data analysis: averages, standard deviation, trends and graphs.
- Artificial Intelligence: prediction, classification and healthcare applications.
- Automation: repeating calculations and data-processing tasks.

3. Basic Components of a Computer

3.1 Input Devices

Devices used to enter data: keyboard, mouse, scanner, microphone and touchscreen.

3.2 Processing Unit

The CPU (Central Processing Unit) processes instructions and data. Major parts include ALU, Control Unit and registers.

3.3 Memory and Storage

Primary memory includes RAM and ROM. Secondary storage includes SSD, hard disk, pen drive and memory card.

3.4 Output Devices

Devices that present results: monitor, printer, speaker and projector.

4. Hardware and Software

Hardware	Software
Physical components	Programs/instructions
Can be seen and touched	Cannot be physically touched
Keyboard, monitor, mouse	Python, Word, Excel
Printer, SSD	Windows, Linux, browser

5. Data and Information

Data means raw facts and figures. Example: 10.2, 11.5, 10.8, 12.1.

Information is processed and meaningful data. Example: Average experimental reading = 11.15.

6. Types of Software

System software: Controls computer hardware; examples: Windows, Linux, Android.

Application software: Performs user tasks; examples: Word, Excel, PowerPoint.

Programming software: Used to create programs; examples: Python, C, C++, Java.

7. Operating System

An Operating System (OS) is system software that manages computer hardware and provides an environment for running applications.

Examples: Windows, Linux, macOS and Android.

Functions include memory management, file management, process management, device management, security and user interface.

8. Files

A file is a collection of related information stored under a name. Examples: pharmacy_data.xlsx, experiment.csv, report.docx and pharmacy_calculation.py. Python programs normally use the .py extension.

PART B – INTRODUCTION TO PROGRAMMING

9. What is a Program?

A program is a set of instructions given to a computer to perform a particular task.

```
price = 50
quantity = 5
total = price * quantity
print(total)
```

Output: 250

10. What is Programming?

Programming is the process of designing and writing instructions that a computer can execute.

General process: Problem → Algorithm/Steps → Program → Execution → Output

11. What is a Programming Language?

A programming language is a language used to communicate instructions to a computer. Examples: Python, C, C++, Java, JavaScript and R.

PART C – INTRODUCTION TO PYTHON

12. What is Python?

Python is a high-level, general-purpose programming language known for its simple and readable syntax. It was created by Guido van Rossum and first released in 1991.

13. Why Learn Python in Pharmacy?

Python is not the only tool used in Pharmacy. Excel, SPSS, R and MATLAB are also useful. Python is valuable because it provides a flexible programming foundation connecting calculations, data analysis, visualization, automation and AI.

- Pharmaceutical calculations – dose, dilution, concentration, percentage and unit conversion.
- Laboratory data analysis – processing experimental observations and calculating results.
- Data visualization – creating graphs and charts for research and laboratory data.
- Pharmacokinetic analysis – studying drug concentration and time-related measurements.
- Drug discovery – molecular data analysis, screening, bioinformatics and computational drug discovery.
- Clinical research – analyzing clinical-trial and patient-related datasets.
- Artificial Intelligence – machine learning and predictive applications in healthcare and drug discovery.
- Automation – repeatedly processing large numbers of calculations or records.

14. Python Compared with Other Tools

Tool	Common Strength
Excel	Tables, calculations and basic data analysis
SPSS	Statistical analysis
R	Statistics and research
MATLAB	Numerical and scientific computing
Python	Programming, data analysis, visualization, automation and AI

Important: Python does not make other tools useless. Different tools have different strengths. Python gives students a flexible programming foundation for modern pharmaceutical research and healthcare applications.

15. Advantages of Python

- Easy-to-read syntax for beginners.
- Open-source programming language.
- Large ecosystem of libraries.
- Works on Windows, Linux and macOS.
- Widely used in scientific computing, data analysis, research and AI.

PART D – FIRST PYTH ON PROGRAM

16. Your First Python Program

```
print("Hello World")
```

Output: Hello World

```
print("Welcome to Python for Pharmacy")
```

17. Understanding print()

The print() function is used to display information on the screen.

```
print("Pharmacy")
print("Python")
print("Drug Analysis")
```

Text is normally written inside quotation marks.

18. Printing Numbers

```
print(100)
print(25)
print(3.14)
```

Numbers can be printed without quotation marks. print(100) is a number; print("100") is text.

19. Variables

A variable is a name used to store a value.

```
age = 20
```

Here, age is the variable and 20 is its value.

```
drug_name = "Paracetamol"
dose = 500
unit = "mg"

print(drug_name)
print(dose)
print(unit)
```

20. Basic Arithmetic Operators

Operator	Meaning	Example
+	Addition	10 + 5
-	Subtraction	10 - 5
*	Multiplication	10 * 5
/	Division	10 / 5

```
a = 20
b = 5

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Output: 25, 15, 100, 4.0

21. Pharmacy Example – Total Tablets

```
tablets_per_strip = 10
number_of_strips = 5
total_tablets = tablets_per_strip * number_of_strips
print("Total tablets =", total_tablets)
```

Output: Total tablets = 50

22. Taking Input

The input() function allows the user to enter information while the program is running.

```
name = input("Enter your name: ")
print("Hello", name)
```

23. Pharmacy Input Example

```
drug = input("Enter drug name: ")
print("Selected drug:", drug)
```

24. Input and Type Conversion

Data entered using input() is initially treated as text. Use int() for whole numbers and float() for decimal numbers.

```
quantity = int(input("Enter quantity: "))
print(quantity * 10)
```

```
price = float(input("Enter price: "))
print(price)
```

25. Complete Pharmacy Program – Total Medicine Cost

```
medicine = input("Enter medicine name: ")
price = float(input("Enter price per strip: "))
quantity = int(input("Enter number of strips: "))

total = price * quantity

print("Medicine:", medicine)
print("Price per strip:", price)
print("Quantity:", quantity)
print("Total cost:", total)
```

Sample Output:

```
Enter medicine name: Paracetamol
Enter price per strip: 25
Enter number of strips: 4
Medicine: Paracetamol
Price per strip: 25.0
Quantity: 4
Total cost: 100.0
```

PART E – CLASSROOM ACTIVITIES

Activity 1: Predict the Output

```
x = 10
y = 5
print(x + y)
print(x * y)
print(x / y)
```

Ask students to predict the output before running the program.

Activity 2: Pharmacy Calculation

One strip contains 10 tablets. A patient needs 7 strips. Write a Python program to calculate the total number of tablets.

```
tablets = 10
strips = 7
total = tablets * strips
print("Total tablets =", total)
```

Activity 3: Medicine Cost

Price = ₹35 and quantity = 8. Ask students to write a program to calculate the total cost. Expected result: ₹280.

PART F – IMPORTANT QUESTIONS

Very Short Questions

1. What is a computer?
2. What is data?
3. What is information?
4. What is hardware?
5. What is software?
6. What is an operating system?
7. What is a program?
8. What is programming?
9. What is Python?
10. Who created Python?
11. What is a variable?
12. What is the use of print()?
13. What is the use of input()?
14. What is the purpose of int()?
15. What is the purpose of float()?

Short Questions

16. Explain the basic components of a computer.
17. Differentiate hardware and software.
18. Explain the importance of computers in Pharmacy.
19. Explain why Python is useful for Pharmacy students.
20. Compare Python with Excel, R and SPSS.
21. Explain variables with a Python example.
22. Explain the input() function.

23. Explain basic arithmetic operators in Python.

Practical Questions

24. Write a Python program to display your name and department.
25. Write a program to calculate total tablets.
26. Write a program to calculate medicine cost.
27. Write a program to calculate the average of three readings.
28. Write a program that accepts a medicine name and price from the user.

Quick Revision

Concept	Meaning
Computer	Electronic device that processes data
Data	Raw facts and values
Information	Processed meaningful data
Hardware	Physical parts of a computer
Software	Programs/instructions
Program	Set of instructions
Python	Programming language
Variable	Stores a value
print()	Displays output
input()	Takes user input
int()	Converts to integer
float()	Converts to decimal number
+ - * /	Basic arithmetic operators

Final Message for Students

You are not learning Python to become software engineers. You are learning Python because modern Pharmacy increasingly depends on computation, data analysis, research, automation, drug discovery and healthcare technology. The goal is to combine Pharmacy knowledge with computational skills.

Recommended Teaching Flow for Class 1

Computer Fundamentals → Why computers in Pharmacy → Why Python → Python vs other tools → First Python program → Variables → Input → Simple Pharmacy calculation → Student activity